

Project Title: Sephora – Beauty E-Commerce Database Management System

DBMS – PAYMENT MANAGEMENT

1. Database Design

1.1 Payment Table

Attribute	Data Type	Key	Description
Payment_ID	INT	PK	Unique payment ID
Order_ID	INT	FK	References Orders
Payment_Method	VARCHAR(30)	-	UPI / Card / COD
Transaction_ID	VARCHAR(100)	-	Transaction ID
Payment_Status	VARCHAR(20)	-	Successful / Failed
Payment_Date	DATE	-	Date of payment
Amount	DECIMAL(10,2)	-	Payment amount

2. Table Relationships

Customer – Orders = 1 : M

Orders – Payment = 1 : 1

Orders – Order_Items = 1 : M

Product – Order_Items = 1 : M

Category – Product = 1 : M

3. Table Creation

3.1 Create Payment Table

```
SQL> CREATE TABLE Payment (  
2     Payment_ID NUMBER(5) PRIMARY KEY,  
3     Order_ID NUMBER(5) NOT NULL,  
4     Payment_Method VARCHAR2(30) NOT NULL,  
5     Transaction_ID VARCHAR2(100) NOT NULL,  
6     Payment_Status VARCHAR2(20) NOT NULL,  
7     Payment_Date DATE NOT NULL,  
8     Amount NUMBER(10,2) NOT NULL,  
9     CONSTRAINT fk_payment_order  
10        FOREIGN KEY (Order_ID)  
11        REFERENCES Orders(Order_ID)  
12 );
```

Table created.

4. Payment Data Insertion

```
SQL> INSERT INTO Payment  
2     (Payment_ID, Order_ID, Payment_Method, Transaction_ID,  
3     Payment_Status, Payment_Date, Amount)  
4     VALUES
```

```

5 (701, 501, 'UPI', 'UPI5012026',
6 'Successful', TO_DATE('10-SEP-2026','DD-MON-YYYY'), 1450.00);

1 row created.

SQL> INSERT INTO Payment
2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
3 Payment_Status, Payment_Date, Amount)
4 VALUES
5 (702, 502, 'Card', 'CARD5022026',
6 'Successful', TO_DATE('16-SEP-2026','DD-MON-YYYY'), 3298.50);

1 row created.

SQL> INSERT INTO Payment
2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
3 Payment_Status, Payment_Date, Amount)
4 VALUES
5 (703, 503, 'UPI', 'UPI5032026',
6 'Successful', TO_DATE('12-SEP-2026','DD-MON-YYYY'), 2199.20);

1 row created.

SQL> INSERT INTO Payment
2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
3 Payment_Status, Payment_Date, Amount)
4 VALUES
5 (704, 504, 'Card', 'CARD5042026',
6 'Successful', TO_DATE('13-SEP-2026','DD-MON-YYYY'), 4899.05);

1 row created.

SQL> INSERT INTO Payment
2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
3 Payment_Status, Payment_Date, Amount)
4 VALUES
5 (705, 505, 'COD', 'COD5052026',
6 'Successful', TO_DATE('14-SEP-2026','DD-MON-YYYY'), 1249.10);

1 row created.

SQL> INSERT INTO Payment
2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
3 Payment_Status, Payment_Date, Amount)
4 VALUES
5 (706, 506, 'UPI', 'UPI5062026',
6 'Failed', TO_DATE('15-SEP-2026','DD-MON-YYYY'), 2362.24);

1 row created.

SQL> COMMIT;

Commit complete.

```

5. Manage Payment Transactions

5.1 Display Successful Payments

```

SQL> SELECT *
2 FROM Payment
3 WHERE Payment_Status = 'Successful';

```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	TRANSACTION_ID	PAYMENT_STATUS	PAYMENT_D	AMOUNT
701	501	UPI	UPI5012026	Successful	10-SEP-26	1450
702	502	Card	CARD5022026	Successful	16-SEP-26	3298.5
703	503	UPI	UPI5032026	Successful	12-SEP-26	2199.2
704	504	Card	CARD5042026	Successful	13-SEP-26	4899.05
705	505	COD	COD5052026	Successful	14-SEP-26	1249.1

```
5 rows selected.
```

5.2 Display Failed Payments

```
SQL> SELECT *
      2 FROM Payment
      3 WHERE Payment_Status = 'Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	TRANSACTION_ID	PAYMENT_STATUS	PAYMENT_D	AMOUNT
706	506	UPI	UPI5062026	Failed	15-SEP-26	2362.24

```
1 row selected.
```

6. Modify Payment Status

```
SQL> UPDATE Payment
      2 SET Payment_Status = 'Successful'
      3 WHERE Payment_ID = 706;
```

```
1 row updated.
```

```
SQL> COMMIT;
```

```
Commit complete.
```

```
SQL> SELECT *
      2 FROM Payment
      3 WHERE Payment_ID = 706;
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	TRANSACTION_ID	PAYMENT_STATUS	PAYMENT_D	AMOUNT
706	506	UPI	UPI5062026	Successful	15-SEP-26	2362.24

7. Analyze Payment Methods

7.1 Calculate Total Transactions by Payment Method

```
SQL> SELECT
      2     Payment_Method,
      3     COUNT(*) AS Total_Transactions
      4 FROM Payment
      5 GROUP BY Payment_Method
      6 ORDER BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_TRANSACTIONS
COD	1
Card	2
UPI	3

7.2 Calculate Amount Collected by Payment Method

```
SQL> SELECT
      2     Payment_Method,
      3     SUM(Amount) AS Total_Amount
      4 FROM Payment
      5 WHERE Payment_Status = 'Successful'
      6 GROUP BY Payment_Method
      7 ORDER BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_AMOUNT
COD	1249.1
Card	8197.55
UPI	6011.44

8. Payment Transaction Report

```
SQL> SELECT
2     p.Payment_ID,
3     o.Order_ID,
4     c.First_Name || ' ' || c.Last_Name AS Customer_Name,
5     pr.Product_Name,
6     cat.Category_Name,
7     p.Payment_Method,
8     p.Transaction_ID,
9     p.Payment_Date,
10    p.Amount,
11    p.Payment_Status
12 FROM Payment p
13 JOIN Orders o
14     ON p.Order_ID = o.Order_ID
15 JOIN Customer c
16     ON o.Customer_ID = c.Customer_ID
17 JOIN Order_Items oi
18     ON o.Order_ID = oi.Order_ID
19 JOIN Product pr
20     ON oi.Product_ID = pr.Product_ID
21 JOIN Category cat
22     ON pr.Category_ID = cat.Category_ID
23 ORDER BY p.Payment_Date DESC;
```

PAYMENT_ID	ORDER_ID	CUSTOMER_NAME	PRODUCT_NAME	CATEGORY_NAME	PAYMENT_METHOD	TRANSACTION_ID	PAYMENT_D	AMOUNT	PAYMENT_STATUS
702	502	Meera Krishnan	Luxe Eyeshadow Palette	Makeup	Card	CARD5022026	16-SEP-26	3298.5	Successful
706	506	Divya Nair	Rose Eau de Parfum	Fragrance	UPI	UPI5062026	15-SEP-26	2362.24	Successful
705	505	Sneha Iyer	Hydrating Moisturizer	Skincare	COD	COD5052026	14-SEP-26	1249.1	Successful
704	504	Nandini Rao	Foundation SPF 30	Makeup	Card	CARD5042026	13-SEP-26	4899.05	Successful
703	503	Kavya Reddy	Vitamin C Serum	Skincare	UPI	UPI5032026	12-SEP-26	2199.2	Successful
701	501	Ananya Sharma	Velvet Matte Lipstick	Makeup	UPI	UPI5012026	10-SEP-26	1450	Successful

6 rows selected.